

SynthDB: Synthesizing Database via Program Analysis for Security Testing of Web Applications

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Testing database-backed web applications is challenging because their behaviors (e.g., control flow) are highly dependent on data returned from SQL queries. Without a database containing sufficient and realistic data, it is challenging to reach potentially vulnerable code snippets, limiting various existing dynamic-based security testing approaches. However, obtaining such a database for testing is difficult in practice as it often contains sensitive information. Sharing it can lead to data leaks and privacy issues.

In this paper, we present SYNTHDB, a program analysis-based database generation technique for database-backed PHP applications. SYNTHDB leverages a concolic execution engine to identify interactions between PHP codebase and the SQL queries. It then collects and solves various constraints to reconstruct a database that can enable exploring uncovered program paths without violating database integrity. Our evaluation results show that the database generated by SYNTHDB outperforms state-of-the-arts database generation techniques in terms of code and query coverage in 17 real-world PHP applications. Specifically, SYNTHDB generated databases achieve 62.9% code and 77.1% query coverages, which are 14.0% and 24.2% more in code and query coverages than the state-of-the-art techniques. Furthermore, our security analysis results show that SYNTHDB effectively aids existing security testing tools: Burp Suite, Wfuzz, and webFuzz. Burp

Suite aided by SYNTHDB detects 76.8% of vulnerabilities while other existing techniques cover 55.7% or fewer. Impressively, with SYNTHDB, Burp Suite discovers 33 previously unknown vulnerabilities from 5 real-world applications.

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